COMPARATIVE BEHAVIOR OF PRIMATES[1]

I. DELAYED REACTION TESTS ON PRIMATES FROM THE LEMUR TO THE ORANG-OUTAN

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I. INTRODUCTION

The purpose of the following investigation was to compare the ability of primates, ranging from the lemur to the orang-outan, on tests of delayed reaction. The delayed reaction method is particularly adaptable for such a project as all members of the order of primates may be tested using a single type of apparatus and a uniform procedure. Differences in capacity may be determined by the maximum delay to which the individual animals can successfully respond, by the degree of perfection of response attained by the various primates at each interval of delay, and by the nature of the learning process in so far as this is indicated by learning curves.

II. HISTORICAL

A. Delayed reaction tests

The first extensive use of the delayed reaction experiment was made by Hunter (17) whose original method was subsequently modified by the same investigator (18) and used in testing a child. Hunter's second and more direct method of attack was subsequently modified by Yerkes (29), (30), who used it to test mnemonic processes in the gorilla and chimpanzee, and by Tinklepaugh (27) in testing representative factors in the macaque.

[1] I wish to thank Mrs. V. A. C. Henmon who suggested the possibility of using the primates in the zoological park as subjects for research and who assisted in the tests on the orang-outan; and to Mr. Fred Winkelman, director of the Madison Zoological Park, whose constant help and advice made possible this research.

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B. Comparison of primate types

From 1889 to 1907, Bianchi (5) recorded observations on the adaptive behavior of 2 baboons (Cynocephalus porcarious) and of 7 cebus monkeys, both before and after partial extirpation of the cortex. He drew no conclusions concerning their respective abilities, however, and his tests were so haphazard and unstandardized that it is impossible for another person to do so.

Haggerty (14) in 1909 tested the comparative ability of two genera of monkeys, 10 cebus and 3 macaques, to solve a series of problem box experiments, either independently or by means of imitation. On the basis of his rather limited data he concluded that there was no outstanding animal or specie and that the results were about even.

The experiments of Nellmann and Trendelenberg (22) and Drescher and Trendelenberg (8) indicate that the monkey and gibbon show about the same level of adaptive ability in drawing food toward themselves with a rake.

Human beings from eight to thirty-four years of age were clearly superior to all other primates in the preservation tests carried out by Hamilton (15), (16), although a twenty-six months old human infant was inferior to even a "defective" monkey. The monkeys tested included 4 macaques and 1 baboon, the performance of the baboon being better than that of three and poorer than that of one of the other monkeys.

A delayed reaction method offering from five to eight choices was used by Revesz (23), (24). Children from four to seven years of age proved to be superior to monkeys in their memory of both the absolute and relative position of the correct container.

Alpert (4) tested 44 children on problems similar to those used by Kohler (19) on 9 chimpanzees and concluded finally that the differences between the two families such as "the absence of solutions with partial insight in the chimpanzee, the absence of delayed solutions among them, and the greater dependence of the chimpanzee upon the physical juxtaposition of objective and tool in order to see their functional relationship . . . . may be explained by the superior mental and emotional development of the child and by differences in technique."

A similar study is that in which Aldrich and Doll (3) studied the ability of 8 idiot children to solve the box stacking problem. They compared their results with the data previously obtained by Bingham (6) who had tested 4 chimpanzees in a situation which was almost identical. Three idiots whose mental age was twenty months or less failed to solve the problem, but 5 others with mental ages ranging from thirty-two to thirty-eight months were successful. Aldrich and Doll concluded that there was little difference in the initial performances of their brighter idiot group and Bingham's chimpanzees, but that "the idiots more quickly perfected their solutions in succeeding trials, both from the standpoint of time and the elimination of fumbling."

The experiments of Gellermann (10), (11), (12), on the double alternation problem show clearly that the human adult and child (over four years old) are quite superior to the Macacus rhesus monkey, in spite of the fact that this animal can solve the double alternation problem with greater facility than any other animal so far tested.

III. CLASSIFICATION OF THE ANIMALS, THEIR HOUSING AND CARE

The primates tested in this investigation included 2 anthropoid apes, an orang-outan and a white-handed gibbon, 19 other Catarrhine, or old-world monkeys, 2 Platyrrhine, or new-world monkeys, and 1 lemur. These animals are listed in table 1 according to suborders, families, genus and species, following the classification of Elliot (9). The familiar name, approximate age at the beginning of the experimental period, and sex are also included.

All of the animals were housed in a modern building especially designed for primates at the Vilas Park Zoological Gardens at Madison, Wisconsin. The larger animals were housed singly and the smaller ones in groups of from 2 to 4 in iron barred cages 84 inches high, 72 inches wide, and either 72 or 144 inches long. The floor of these was of wood and was 29 inches above the floor of the building. Each of the cages was provided with a swinging bar, a steel chain with a wooden block attached to the end, and a half floor. All of these indoor cages led by means of a narrow

TABLE 1

Classification of the animals

FAMILY	GENUS	SPECIES	FAMILIAR NAME	ESTIMATED AGE	SEX
Suborder Anthropoidea					
				years	
Pongiidae (orang-outan)	Pongo	Pygmaeus	Jiggs	11½	Male
Hylobatidae (white-handed gibbon)	Hylobates	Lar	Charlotte	15	Male
Lasiopygidae (Macacus rhesus)	Pithecus	Rhesus	No. I	7	Male
Lasiopygidae (Macacus rhesus)	Pithecus	Rhesus	No. II	5	Female
Lasiopygidae (Macacus rhesus)	Pithecus	Rhesus	No. III	5	Female
Lasiopygidae (Macacus rhesus)	Pithecus	Rhesus	No. IV	5	Female
Lasiopygidae (Macacus rhesus)	Pithecus	Rhesus	No. V	4	Female
Lasiopygidae (Java monkey)	Pithecus	Mordax	Sourface	5	Male
Lasiopygidae (Java monkey)	Pithecus	Mordax	Kewpie	5	Male
Lasiopygidae (Barbary ape)	Simia*	Sylvanus	Blackie	5	Male
Lasiopygidae (sooty mangaby)	Cercocebus	Aethiops	Sooty	3	Male
Lasiopygidae (Mona guenon)	Lasiopyga	Mona	Mona	3	Female
Lasiopygidae (green guenon)	Lasiopyga	Callitrichus	Thunder	9	Male
Lasiopygidae (green guenon)	Lasiopyga	Callitrichus	Lightning	9	Male
Lasiopygidae (mustached guenon)	Lasiopyga	Cephus	Skeezix	1½	Male
Lasiopygidae (yellow baboon)	Papio	Cynocephalus	Tommy	7	Male
Lasiopygidae (guinea baboon)	Papio	Papio	Andy	7	Male
Lasiopygidae (guinea baboon)	Papio	Papio	Min	7	Female
Lasiopygidae (mandrill)	Papio	Sphinx	Pat	20	Male
Lasiopygidae (mandrill)	Papio	Sphinx	Socrates	1½	Male
Lasiopygidae (mandrill)	Papio	Sphinx	Xanthippe	1½	Female
Cebidae (Capuchin monkey)	Cebus	Capucinus	Scarface Al	4	Male
Cebidae (Capuchin monkey)	Cebus	Capucinus	Bugs	4	Male
Suborder Lemuroidea					
Nycticibidae	Lemur	Variegatus	Racket	13	Male

* The Barbary ape is classified by Elliot as a separate genus, Simia; by Sonntag (25) it is classified as belonging to the genus Macacus (Elliot's genus Pithecus). We include it with the macaques in the discussion of results.

alley-way at the top of the cage to an outdoor cage of approximately equal size, where the animal might go when the weather permitted. This passageway could be blocked by means of a sliding door which made it possible to restrain any desired animal within the inside cage, and to segregate them when several were kept in a single cage.

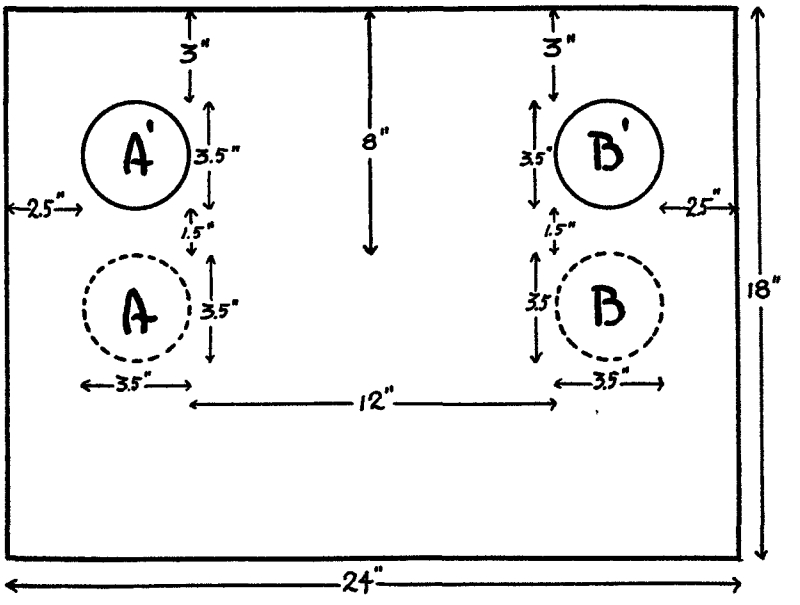
The primates were cared for by the attendants at the park. The cages were cleaned each morning at about 8:30 and the animals fed twice a day at 9:00 a.m. and 4:00 p.m. on a vegetable diet. Water was provided by automatic fountains at all times. During the winter the animals were provided each day with a quantity of clean straw.

IV. APPARATUS

A. Direct method

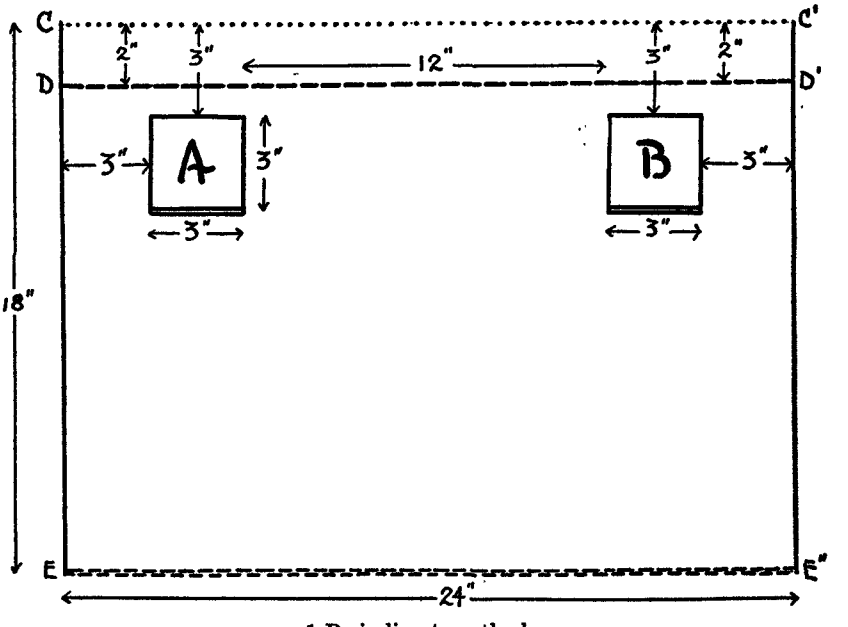
The apparatus used in the direct method of testing delayed reactions consisted of a table and the appropriate containers. The table was painted black and stood 29 inches high, thereby being level with the floor of the cages.

Two white enameled tin cups $2\frac{1}{2}$ inches high and $3\frac{1}{2}$ inches in diameter with the handles removed were used as containers, save in the cases of the cebus monkeys and the lemur. For these animals metal containers 3 inches high and $1\frac{3}{4}$ inches in diameter were substituted as they were more easily displaced than the tin cups. Originally all the cups were placed free upon the surface of the table, but later they were fastened at the back to a 3-inch string secured at its other end to a screw in the table top. This prevented the cups from being knocked on the floor and as a result frightening the monkeys and inconveniencing the experimenter. The containers were so placed that their inner surfaces were in every case 12 inches apart. At the time of choice and during the period of delay the front of the containers was 3 inches from the front of the table. Before being used to cover the food, they were placed 5 inches farther back. The dimensions of the surface of the table and the positions of the containers are given in figure 1a. The experimenter stood directly behind the table at all times.



1-A, direct method

A and *B*, position of containers before placing food. *A'* and *B'*, position of containers during delay.



1-B, indirect method

A–B, stationary containers in indirect method. *C–C'*, restraining screen. *D–D'*, concealing screen. *E–E'*, one-way visual screen.

FIG. 1. DIAGRAM OF FLOOR PLAN OF DELAYED REACTION APPARATUS

B. Indirect method

The apparatus used in the indirect method of testing delayed reactions was like the first, save for the fact that the top of the table, which was 24 by 18 inches wide, formed the base of a metal box with side walls 38 inches high. Inside this box were three screens. The first screen, the restraining screen, was flush with the front of the box and was made of extra heavy ½-inch wire screening. This kept the monkey from reaching to the box but did not prevent it from seeing the container under which the food was placed. The second screen, the concealing screen, placed 2 inches behind the first, was made of heavy black cloth and when lowered it entirely hid the containers. At the back of the box was a one-way visual screen. The first two screens were 20 inches high, the third screen 32 inches. All the screens could be raised and lowered as desired. Metal containers, 3 by 3 by 3 inches, were placed 1 inch behind the second screen and fastened by hinges in the back. A small wire running from the inside of the cup to the experimenter made it possible to hold the cup to the table top when desired. A diagram of the floor plan of the apparatus is given in figure 1b.

V. PROCEDURE

Preliminary training

All of the primates were given a period of preliminary training to accustom them to the general setting of the problem and also to give them opportunity to learn to push over the container and obtain food. In these trials the table was placed out of reach of the animal and a single piece of food was placed in the middle of the table top, 3 inches from the front. The food was then covered by a container, the table was brought forward even with the front of the cage, and the primate was permitted to push over the cup and obtain the reward. In the early trials most of the monkeys experienced some difficulty in lifting up the containers and often resorted to the device of simply knocking them off the table, an unsatisfactory response as the food was also frequently dislodged. After the period of from 20 to 50 trials, however, these difficulties

usually disappeared. Most of the animals appeared to solve immediately the problem involved in removing the cup to obtain the food, a task which Gesell (13) places at about the nine months level of human development. The young mustached guenon failed to solve this problem even when glass containers were substituted for metal ones as was suggested by Tinklepaugh (27, p. 210). The lemur failed to respond to the container for about three months, from either inability to grasp the problem or simple indifference to the task, but was successful later. The 2 cebus monkeys were extremely timid, but appeared to solve this preliminary task immediately after becoming accustomed to the general experimental situation.

General procedure

A. Direct method. As soon as the animal had become accustomed to the general nature of the problem and had learned to displace the cup with reasonable skill, the test series of delayed reaction experiments was begun. In these tests food was placed in one of the two front corners of the table and the two containers were then simultaneously placed in their proper positions, one of them over the food and one empty. The table was immediately brought forward to the front of the cage and the animal allowed to make its choice. If the primate was successful, the table was withdrawn as soon as the food was taken; if not, it was removed when he knocked over the incorrect container. In either case the second container was then removed, and if the animal had erred, he was permitted to see the unobtained reward. The food was then removed and the procedure repeated. To prevent the formation of direction habits the order of presentation was varied at random but in such manner that not more than two alternate or three successive choices on a single side were possible. The period of delay between choices varied from fifteen to one hundred and twenty seconds but was otherwise not controlled. As is suggested in the researches of Adams (1) on cats, this may have influenced our results, particularly on the shorter delays. In the shortest delays (recorded as zero seconds) the table was pushed forward as soon as the cups were placed, so that in reality a delay

of from one to two seconds actually occurred. Since a similar method of recording was followed throughout, all of the delays are slightly longer than the period given. The first 2 trials of each day's work were discarded, the animal being given these to allow him to become oriented to the experimental situation. From 20 to 40 trials per day were then given to the animal, depending on the interest and motivation that the particular primate showed toward the task. All the animals were given from fifty- to one hundred- of the zero-second delays before being tested on the longer delay periods, and when these were introduced they were preceded each day by five to ten of the minimal delays. The longer delays were then given to the animal, 10 trials being allowed for each delay period until the problem became too difficult for the animal, a fact usually indicated not only by a marked increase in the number of errors but also by definite emotional reactions on the part of the monkeys, such as refusal to respond, striking at the cups or table, and snarling or striking at the experimenter. If the coöperation of the monkey could still be obtained he was retested on the series. Placing the longer delays at the end of the series involved the serious danger that they would be given to the animal after his motivation had passed the maximum. Nevertheless, better results were obtained by this method than by introducing the long delays in the early trials, save possibly in the case of very well trained and emotionally stable monkeys. After the monkeys failed on the longer delays (often with accompanying emotional reactions), they were almost always still highly motivated for shorter delays, indicating that their previous mistakes were certainly not caused by insufficient motivation alone.

At the beginning of this research the animals were tested three times a week, Monday, Wednesday, and Friday from 1:00 p.m. to 3:30 p.m. Later the animals were tested on all days except Saturday and Sunday, and the testing was done from 10:00 a.m. to 12:30 p.m. It was not always possible to test a single animal on successive days as the tests demanded the coöperation of the primates. This, however, was done in so far as it was possible. Malaga grapes cut in two and small pieces of oranges were used to supply a suitable incentive.

B. Indirect method. With the second or indirect type of apparatus the restraining and concealing screens were first lowered, the containers drawn back on their hinges so that the table areas they normally covered were exposed, and the food was then placed in the center of one of these areas. The concealing screen was then raised enabling the primate to see the food while the restraining screen, which was left lowered, prevented him from obtaining it. As soon as the experimenter was assured that the animal has seen the food, the containers and then the concealing screen were lowered for the period of delay, thus hiding the entire table top. At the conclusion of the delay period, the concealing screen was first raised, the restraining screen was raised immediately afterward, and the animal permitted to make its choice. In case of error the monkey was prevented from obtaining food as it was possible to hold the containers to the floor by means of a wire leading from the inner edge of the container, under the floor of the table, to the experimenter.

VI. RESULTS

A. Direct method of testing delayed reaction

In table 2 are recorded the number of trials given to each animal at periods of delay ranging from zero to six hundred seconds, the per cent of the total trials solved correctly, and the per cent of the last 50 trials correct. In figure 2 are recorded the learning curves of the animals responding to delays of zero and thirty seconds. For this purpose the trials in groups of 25 were plotted along the abscissa and the number of errors made in each of these 25 trials was plotted along the ordinate.

Comparison of the various genera. In this investigation it was possible to test only one higher anthropoid ape, the orang-outan "Jiggs" and he unfortunately died less than a month after the experiment was started. There is every reason to believe, however, that his level of performance was definitely above that of the other primates of our group. His learning curves indicate immediate grasp of the problem for the zero- to thirty-second delays, and on delays of six hundred seconds he showed definite ability to solve the problem with little practice.

TABLE 2

Summary of correct responses

ANIMAL	FAMILIAR NAME	PERIOD OF DELAY																							
		0 second			5 seconds			15 seconds			30 seconds			60 seconds			120 seconds			180 seconds			Over 180 seconds		
		NT	PC	LFC	NT	PC	LFC	NT	PC	LFC	NT	PC	LFC	NT	PC	LFC	NT	PC	LFC	NT	PC	LFC	NT	PC	LFC
			per cent	per cent		per cent	per cent		per cent	per cent		per cent	per cent		per cent	per cent		per cent	per cent		per cent	per cent		per cent	
Orang-outan	Jiggs	60	100		60	98		60	100		40	95					20	90					16	88	*
Gibbon	Charlotte	200	89	98	200	90	96	200	86	92	200	80	90	65	71		35	66							
Macaque	No. I	200	94	98	200	94	96	200	90	94	200	92	94	100	82	90	60	75							
Macaque	No. II	200	86	94	100	93	92	100	92	94	100	89	90	60	63		40	65							
Macaque	No. III	200	72	88	200	77	84	200	71	74	200	62	70	100	51	53	60	47							
Macaque	No. IV	100	78	92	100	81	86	100	77	82	60	65		60	55										
Macaque	No. V	200	89	96	200	91	94	200	87	94	200	78	86	100	70	78	100	65	74						
Macaque	Sourface	200	90	100	200	96	98	200	86	96	200	85	98	100	89	96	80	80	86						
Macaque	Kewpie	200	92	98	200	94	100	200	95	94	200	93	96	100	94	96	100	86	90	20	70				
Barbary ape	Blackie	200	92	98	200	98	100	200	90	94	200	89	92	100	70	80	60	73							
Guenon	Thunder	200	82	88	100	86	90	80	78		60	68		20	70										
Guenon	Lightning	100	84	94	100	88	90	100	84	84	60	72		30	73										
Guenon	Mona	200	80	90	200	77	92	200	81	88	200	76	92	100	74	84	80	64							
Guenon	Skeezix	(Failed to grasp problem)																							
Mangabey	Sooty	200	91	94	140	84	88	80	78		60	63		40	55										
Baboon	Tommy	200	99	100	200	100	100	200	99	100	200	93	98	100	90	92	100	84	88						
Baboon	Andy	200	96	98	200	95	98	200	90	96	200	91	94	100	84	88	100	78	80						
Baboon	Min	200	96	100	200	98	98	200	96	100	200	96	98	100	90	94	100	81	84						
Mandrill	Pat	200	96	100	200	99	100	200	96	98	200	90	94	100	92	94	100	84	86	60	80		20	85	†
Mandrill	Socrates	60	73		40	68																			
Mandrill	Xanthippe	200	80	96	100	82	92	100	74	84	70	59													
Cebus monkey	Scarface Al	140	74	78	110	77		30	73																
Cebus monkey	Bugs	90	77		80	76		20	65																
Lemur	Racket	80	89		40	80		40	78		14	57													

NT, number of trials. PC, per cent total trials correct. LFC, per cent last 50 trials correct.

* Three hundred seconds.

† Six hundred seconds.

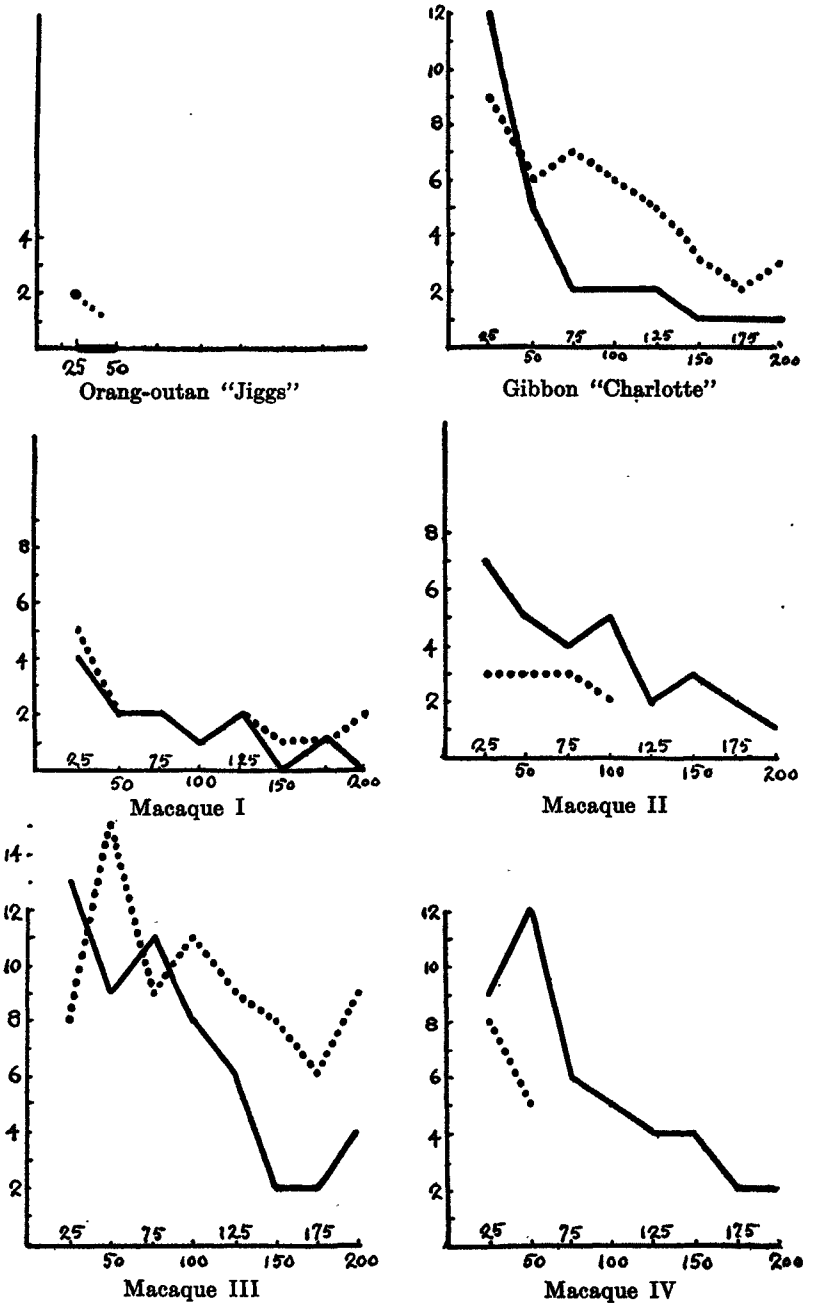
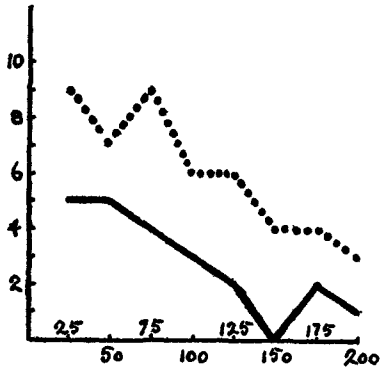
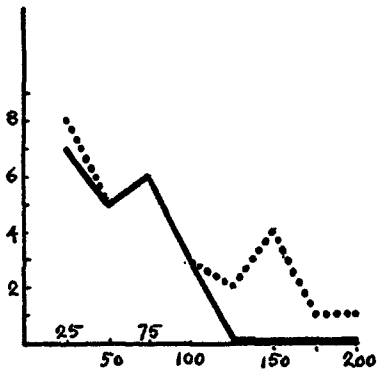


FIG. 2. LEARNING CURVES OF PRIMATES

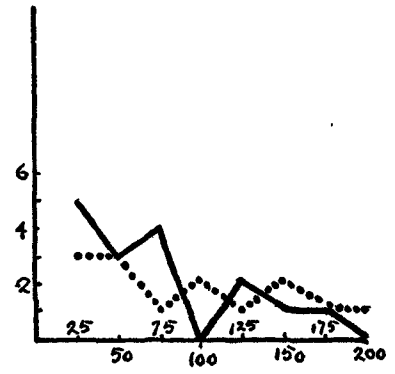
——— 0-second delay. ..... 30-second delay



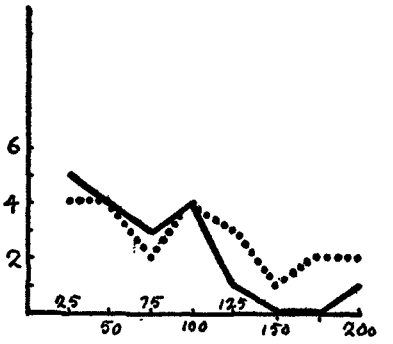
Macaque V



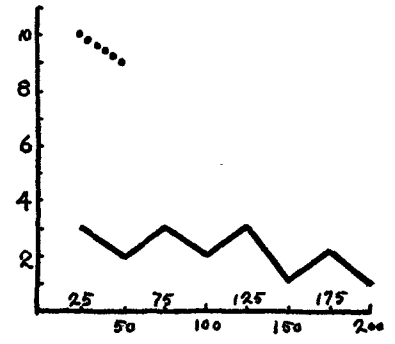
Macaque "Sourface"



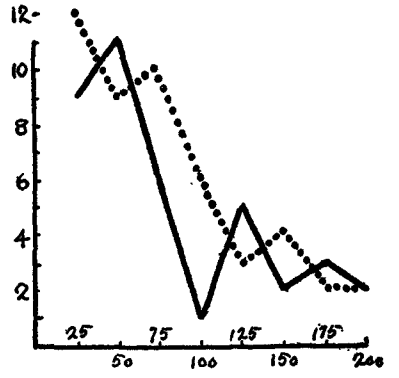
Macaque "Kewpie"



Barbary Ape "Blackie"



Mangabey "Sooty"



Guenon "Mona"

FIG. 2.—*Continued*

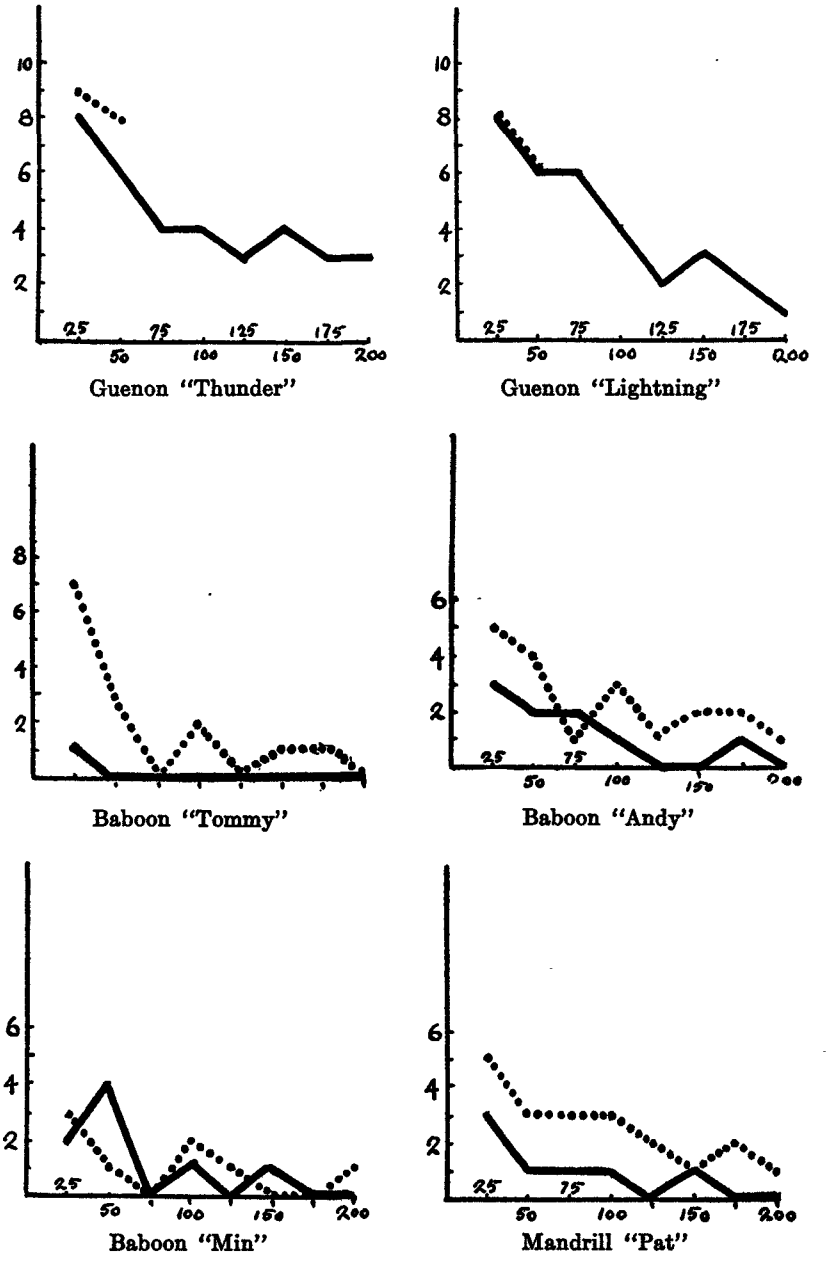
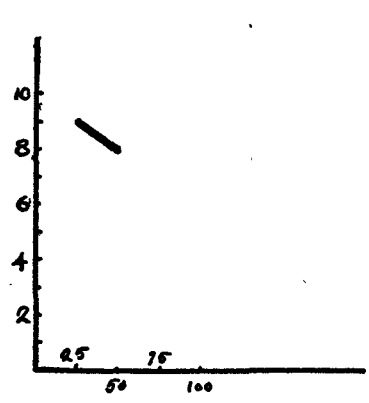
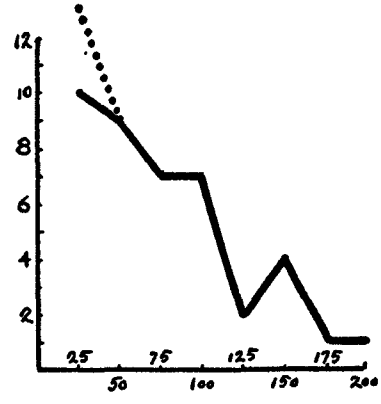


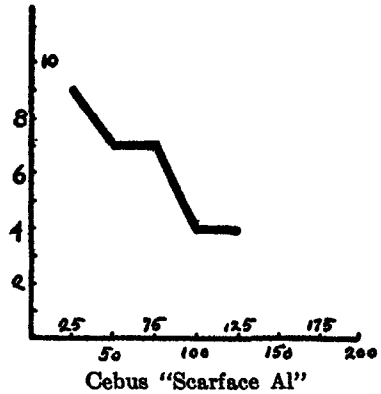
FIG. 2.—*Continued*



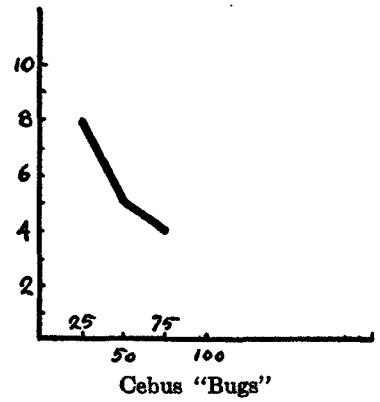
Mandrill "Socrates"



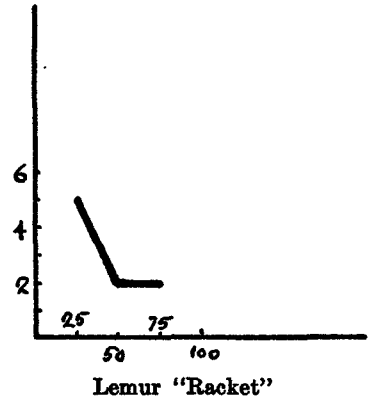
Mandrill "Xanthippe"



Cebus "Scarface Al"



Cebus "Bugs"



Lemur "Racket"

FIG. 2.—*Concluded*

"Charlotte" a white-handed gibbon, classed by many taxonomists as a true anthropoid, gave responses definitely inferior to all of the adult baboons, and his performances were only superior to those of 2 of the 8 macaques. It was quite difficult to obtain responses in this animal for delays sixty seconds or more in length and the responses obtained showed an accuracy in the neighborhood of only about 70 per cent. Conceivably these errors may have been caused by lack of interest and motivation rather than by any intellectual deficiency, as our animal, although perfectly tame and tractable, evinced frequently a total indifference to the entire experimental situation.

Eight macaques, representing three different species, were tested and their responses indicated considerable variation in capacity. Both of the Pithecus mordax (Java monkeys) and one rhesus, number I, passed the criterion of 75 per cent accuracy on delays of one hundred and twenty seconds. The Barbary ape and two of the rhesus, numbers II and V, were 65 per cent accurate or more on the two-minute delays. The remaining two, numbers III and IV, encountered difficulties on delays thirty seconds in length and their responses on the sixty-second delays were little better than chance. The Java monkey, Kewpie, was the only macaque to give evidence of immediate or insight learning, as can be seen by the learning curves of figure 2. The learning curves of macaque II and IV indicate an extremely slow and painful process of trial and error learning even on the zero-second delays.

The baboons, considered to be the most primitive of the Catarrhine monkeys (see Elliot, ii, p. 116) because of their dog-like body build, proved to be remarkably capable performers. All four adults tested surpassed the criterion of 75 per cent correct solutions on one hundred and twenty-second delays, and, without exception, their learning curves indicate a quick grasp of the problem. These performances are at least as good as the best of the performances of the macaques and on the basis of the evidence we have obtained, it would appear that the baboons (genus Papio) are superior both to the macaques (genus Pithecus) and the gibbon (genus Hylobates).

The inferior records of the mandrills, Socrates and Xanthippe, are very probably a result of their immaturity as they were only about a year and a half old at the time of the tests. This period, probably is less than 10 per cent of the normal life span of the animal and less than half the period from birth to adolescence.

Four animals of the genus Lasiopyga were tested, 2 green guenons, 1 Mona guenon, and 1 mustached guenon. The Mona guenon was definitely the best performer of this group, responding adequately (74 per cent) to delays of sixty seconds. Considerable difficulty was found in inducing the green guenons to respond to delays thirty seconds and longer and their accuracy was only about 70 per cent at this period of delay. The mustached guenon, probably less than a year and a half old, failed to solve the problem involved in tipping over the cups and obtaining the food.

One animal of the genus Cercocebus, a sooty Mangabey, was given a short series of tests. This monkey did very well on zero- and five-second delays, grasping the problem quickly and maintaining a high degree of efficiency of response. Incomplete tests on longer delay intervals, however, gave decidedly inferior results.

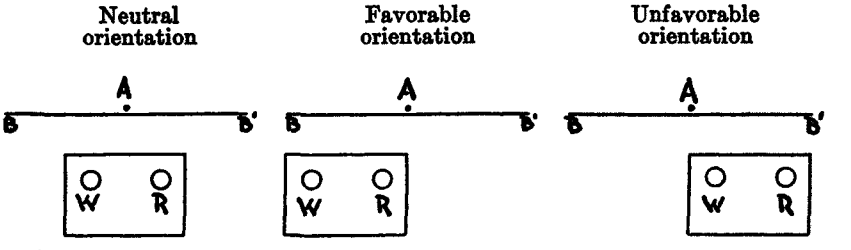
The performances of the 2 cebus monkeys were definitely inferior to those of any of the Lasopygidae. The cebus were extremely timid and shy and were quite difficult to test. On short delays (zero and five seconds) they attained a maximum accuracy of only about 80 per cent and they refused without exception to attempt delays of thirty seconds or more.

The capacity of one lemur, "Racket," was investigated. For two months the animal either would not or could not learn to push over the cups. At the end of this time, however, he suddenly mastered the problem and about a month later began to show definite choice responses. The results (see table 2) indicate that this lemur finally was able to respond successfully to fifteen-second delays.

Bodily orientation. The rôle of bodily orientation in delayed response tests on monkeys has already been discussed by Tinklepaugh, and the results that we obtained are entirely confirmatory of his. Even in the short periods of delay, orientation of the

head was usually lost and the same frequently held true for orientation of the body. In the delays of thirty seconds and over, all bodily orientation was almost invariably destroyed, the animal customarily pacing around the cage during the enforced delay period. Frequently he resumed orientation in the general region of the table before making his choice, but the orientation at this time favored the incorrect container as frequently as the correct one, even when a correct choice was to be subsequently made. Often the animal would start to reach for the wrong container, then check itself and make the correct response.

In order to check the effect of direct bodily orientation on the efficiency of the response, a brief experiment was conducted on the monkeys in which the cups were presented to them in three



A, primate. *B–B'*, front of cage. *R*, correct container. *W*, incorrect container.

FIG. 3. THREE APPROACHES FOR TESTING EFFECT OF BODILY ORIENTATION

different positions following thirty-second delays. A third of the time the table was brought forward immediately in front of the animal so that the correct and incorrect containers were an equal distance from the midline of the monkey, a third of the time the table was brought forward in such a manner that the correct cup was closer to the animal, and a third of the time the table was placed in such a position that the incorrect cup was closer to the primate. These different approaches are illustrated in figure 3, the right-hand cup in these illustrations always being the correct one.

In table 3 is given the percentage of correct responses made in 50 trials by each of the 10 animals in the three situations. The results are quite clear and indicate that if the animal had mastered the problem, its responses were not affected by the presentation

of the wrong cup immediately in front of its head; if the problem had not been mastered, the animal usually selected the closest cup, i.e., the one favored by mere bodily orientation, and if the problem had been only partially mastered, as in the cases of animals V, VI, VII, and VIII, there was a definite tendency for the ratio of errors to increase. This may result from inability of the animal to reorient itself or possibly is merely a function of the distraction occasioned by sudden shifting of the table. The fact that practically no errors are made by any primate when the

TABLE 3

Responses of 10 primates to neutral orientation, favorable orientation, and unfavorable orientation of correct container following 30-second delay periods

Percentages of responses correct in 50 trials

ANIMAL	NEUTRAL ORIENTATION	UNFAVORABLE ORIENTATION	FAVORABLE ORIENTATION
I. Mandrill	100	100	100
II. Pat	98	100	100
II. Baboon Min			
III. Baboon Andy	96	94	98
IV. Baboon Tommy	100	98	100
V. Gibbon Charlotte	90	82	94
VI. Guenon Mona	88	64	100
VII. Macaque No. III	74	32	100
VIII. Guenon Thunder	62	24	96
IX. Mandrill Socrates	54	12	98
X. Macaque No. IV	56	4	100

correct cup is placed nearest the animal makes it appear that the first of these explanations is correct.

Factors determining choice. Solution of the problem of delayed reactions demands that during the delay period the animal maintain some representative factor that stands for the position of the stimulus in space and that its choice be determined by this factor rather than by any other. Certain of the primates appear to have enough ability to meet both of these requirements on simple delays of short duration (up to about thirty seconds) without undergoing an extensive period of training. Other of the primates fail in spite of extended training to solve the simple problem in-

volved in these delays. The majority, however, do learn the problem but only after a training period from 100 to 200 trials in length. This period of training appears to be spent in the elimination of false methods of response, types of responses which are probably more simple biologically; responses which are apparently eliminated by the superior animals without the necessity of any overt activity.

Two types of response are found in which the animal attempts the solution of the problem without any reference to its temporal aspect; in other words, the primate attempts to solve the problem without reference to its own previous responses or to the position in which the food was placed. These responses which may be classified as spatially determined responses involve either: (*a*) reaching to nearest cup, or (*b*) persistent reaching to right or left cup.

(a) Reaching to nearest cup. Having grasped the principle in the preliminary training series that food is in a cup, the simplest possible type of response involving this principle is to reach for the closest cup. The general impression of the experimenters was that this almost invariably occurred in the early trials of those animals that failed to solve the problem by insight. A second response, belonging apparently to the same category, is one in which the primate chooses not the spatially closest cup but the one toward which he is oriented. Experimental evidence, indicating the importance of this basic factor where the problem has not been completely solved, is seen most clearly in the performances of animals VII, VIII, IX, and X in table 3, who favored the closest cup.

(b) Reaching for right or left cup. This type of error was not as common as the first, but still appeared quite frequently during the training period of the animals. In those cases in which it did appear, this habit showed a strong tendency to persist in spite of any methods taken to effect its elimination; the monkey might continue a right- or left-sided position habit even after failing to obtain a food reward for 20 to 30 consecutive trials.

Spatially determined responses to the delayed reaction situation tend to be eliminated in the early part of the learning process

since they constitute an extremely simple type of error, and to give place in turn to temporally determined responses, which may be classified into the following three categories: (*a*) repetition of a previously correct response; (*b*) incorrect response resulting from loss of representative factor during period of delay, and (*c*) correct response determined by representative factor maintained during the period of delay.

(a) Repetition of a previously correct response appears as a very common type of error in delayed reaction experiments, and has been reported in similar situations by Yerkes and Yerkes (30), Hamilton (16), Kuroda (20) and others. Even after this type of error had been eliminated for delays of a certain length, it would frequently appear when the delay period was increased. Experimental verification of the importance of this type of error was obtained by presenting food from the same position four times in succession and then shifting the food on the fifth trial. Repeating this procedure it was possible to determine the percentage of successful responses made on every fifth (critical) trial and to compare this with the percentage of correct responses made by the same animal on the same delay interval when the trials followed the normal series. The results obtained are given in table 4 and indicate that little differences were obtained in the two situations when the period of delay was short enough to be easily within the ability of the monkey, but the four previously correct responses greatly influenced the fifth choice when the period of delay was long enough to seriously test the capacity of the animal.

The existence of the second class of temporally determined responses, incorrect responses resulting from loss of representative factors during period of delay, cannot be shown directly. Since the response of the primates is determined by representative factors, and since the ability to respond successfully decreases as the period of delay is increased, it is logical to assume that many of the responses to the longer delays must have been made after the representative factor had waned or had been lost through retroactive inhibition. Other indirect evidence for the existence of this type of response was seen in the fact that during

TABLE 4

Effect of previous responses

Per cent of 10 trials correct

ANIMAL	15-SECOND DELAY		30-SECOND DELAY		60-SECOND DELAY	
	NS	ES	NS	ES	NS	ES
I. Mandrill (Pat)	100	100	100	80	100	60
II. Baboon (Min)	100	100	100	100	100	70
III. Macaque (No. I)	100	80	90	70	90	40
IV. Guenon (Mona)	100	90	100	40	90	30
V. Macaque (No. IV)	70	20	70	0	NR	NR
VI. Mandrill (Xanthippe)	90	40	80	20	NR	NR

NS, normal or control series; random order of right and left choices. ES, experimental series, food is presented on one side four times in succession before being shifted to the other for the fifth (critical) trial. Results on only critical trials are recorded. NR, no response.

TABLE 5

Tests of retention of delayed reaction method in primates

ANIMAL	ORIGINAL TEST SCORE	RETEST SCORE 10–14 DAYS LATER	SCORE AFTER 4–6 WEEKS OF NO PRACTICE	SCORE AFTER 9–12 WEEKS OF NO PRACTICE
Mandrill (Pat)	58	59	58	56
Baboon (Min)	60	57	58	55
Baboon (Tommy)	57	57	42	49
Macacus (Kewpie)	49	56	51	46
Baboon (Andy)	53	53	47	49
Macacus (Sourface)	51	54	38	51
Gibbon (Charlotte)	50	52	51	49
Barbary ape (Blackie)	50	49	46	47
Macacus (No. I)	42	44	44	37
Guenon (Mona)	42	44	36	33
Macacus (No. II)	40	38	40	32
Mangabey (Sooty)	34	28		
Macacus (No. III)	36	31	33	
Rhesus (No. IV)	29	32	28	
Guenon (Lightning)	33	30	24	28
Guenon (Thunder)	32	24		
Mandrill (Xanthippe)	29	26		
Mandrill (Socrates)	18	18		
Cebus (Scarface Al)	24	22	14	9
Cebus (Bugs)	18	21	14	16
Lemur (Racket)	21	14	16	11

the longer delays the monkeys would frequently reorient themselves and attempt to reach the correct container. Failing in this they would leave and then subsequently choose the wrong cup at the expiration of the predetermined delay period.

The existence of the third type of temporally determined response, correct responses dependent on representative factors maintained during the period of delay, may be inferred from the facts that the percentages of correct responses were in most cases well above chance, and that responses of this type could only be made on the basis of representative factors.

Reliability. An attempt was made to determine the statistical reliability of the delayed reaction technique as employed here. For this purpose a test series of ten trials on each of the delay periods from zero to one hundred and twenty seconds was administered to all of the primates studied, with the exception of the orang-outan, the sooty mangabey and the mustachio monkey. A second series of the same length was given these animals after a period of seven to ten days (the exact time being determined by the temperament of the animal). The total number of correct responses on the first series was correlated with the number correct on the second; this gives a correlation coefficient of .971.[2] We may call this the retest reliability of our battery of tests.

It is admitted that the reliability of the battery is somewhat spuriously high because, in this case, some of the levels were completely failed on both series by the inferior animals.[3] On the other hand, a correlation of this impressive size argues for a high reliability even if this disturbing factor were corrected by separate coefficients for each delay period.

It was also considered important that the retention of the problem be tested. After four to six weeks of no practice, the animals which were available were retested. The total score on this

[2] The formula used in the above correlations was Hull's product-moment formula:

$$\surd = \frac{Mab - MaMb}{\sqrt{M(a^2) - (Ma)^2}\sqrt{M(b^2) - (Mb)^2}}$$

[3] The scores on each test are summarized in table 5.

series correlated .936 with the total score on the series immediately preceding this period of no practice. A second retention test was given after nine to twelve weeks without practice, and the total score for this series correlated .649 with the last series before the period of forgetting. The number of cases in the group tested after seven to ten days was 21; in the four- to six-week group, 17; and in the nine- to twelve-week group, 15.

From these correlations we may reasonably infer: (*a*) that the performance of a primate on this battery of tests may be predicted from one week to the next with great accuracy; (*b*) that the performance after a forgetting period of four weeks is still very closely similar to the original, and (*c*) that only after a period of forgetting of approximately ten weeks is there any great tendency for individual levels of performance to vary.

Effect of age. The number of animals of any genus is so small that only tentative conclusions concerning the effect of age on the delayed reaction tests can be made. It is interesting to note, however, that the mustached guenon, probably less than a year and a half old, was the only one to fail to grasp the problem, and that the two young mandrills did very much poorer than any of the older mandrills. These results are in keeping with those subsequently obtained by Maslow (21) at the Bronx Park Zoo. Although it seems probable from these results that very young animals are inferior in these tests to the more mature monkeys, there is no evidence that the capacity for response diminishes with the onset of old age. The responses of "Racket," the fifteen-year-old lemur of the Vilas Park Zoo, are superior to those of any of the lemurs tested by Maslow, and the responses of the twenty-year-old Mandrill, Pat, are as good as those of any monkey so far tested.

B. Indirect method of testing delayed reactions

The indirect method of testing delayed reactions was not particularly well adapted for testing under conditions such as are found in a city zoo. With the experimenter removed from the situation it was much more difficult to obtain the coöperation of

the animal and to be certain that it had seen the food placed beneath the proper container. The bulky apparatus distracted the more timid monkeys, making it difficult to obtain responses from them, and the time taken in the operation of the doors made it impossible to test the animals on delays less than fifteen seconds long.

In table 6 are given the results obtained on 10 animals. Twenty trials were given to each of 10 animals, using both the direct and

TABLE 6

Comparison of direct and indirect methods

Per cent of correct responses made in 20 trials by 10 primates at delay intervals of fifteen, thirty, and sixty seconds using both methods

ANIMAL	15-SECOND DELAY		30-SECOND DELAY		60-SECOND DELAY	
	DM	IM	DM	IM	DM	IM
I. Mandrill (Pat)	100	100	100	100	100	100
II. Baboon (Min)	100	100	100	90	95	85
III. Macaque (Kewpie)	100	90	100	65	90	NR
IV. Baboon (Tommy)	100	100	100	100	90	95
V. Macaque (No. I)	100	90	100	75	90	70
VI. Macaque (Blackie)	90	60	95	65	75	45
VII. Gibbon (Charlotte)	95	90	85	90	80	80
VIII. Guenon (Thunder)	80	NR	70	NR	NR	NR
IX. Guenon (Mona)	100	NR	90	NR	75	NR
X. Cebus (Bugs)	65	NR	NR	NR	NR	NR

DM, direct method. IM, indirect method. NR, animal would not respond.

indirect methods at delay intervals of fifteen, thirty and sixty seconds. The timidity of the 2 guenons and the cebus monkey made it impossible to obtain any responses from them using the indirect method. The macaque "Kewpie," a timid animal, was badly inhibited by the more complicated apparatus but responded on all but the minute delays. The other monkeys responded almost as successfully to the indirect method as to the direct method, showing very clearly that their responses were not determined by secondary cues derived from the experimenter.

VII. DISCUSSION

Methods

The greater part of the data of this investigation were obtained using a direct method of approach, so there was a theoretical possibility that the animal could make his choices on the basis of extraneous cues. Tinklepaugh has already given this problem careful attention and suggested certain checks to test such a possibility. To guard against the danger of the animal's responses being determined by olfactory cues the following three checks were made: (*a*) food was placed under one container, a screen was interposed, and food was then secretly placed under the second container; (*b*) the cups were hidden by a screen during the delay period and the food shifted from the first to the second container, and (*c*) the entire table top was smeared with fruit juice so that all areas of the table presented an equal olfactory stimulus. The monkeys consistently responded to the cup under which they had seen food placed and were apparently indifferent to alterations of olfactory cues.

The possibility of cues being obtained from the experimenter was tested by: (*a*) having others place the food while the experimenter looked the other way; (*b*) deliberately looking at the wrong container, and (*c*) placing the food and then turning away while the animal made its choice. No evidence was obtained that might indicate that the primate was responding to cues derived from the experimenter. These results are entirely in accord with those obtained by Tinklepaugh. The fact that the monkeys consistently failed when the problem was made difficult also indicated that the response is not made on the basis of extraneous cues. Although the results obtained in this investigation using an indirect method of testing are not entirely satisfactory, they indicate clearly that the monkey is able to make responses of the same order when the experimenter is entirely hidden. Under laboratory conditions where it is possible to control conditions of motivation and distraction, which appeared to hamper the more excitable animals, it should be possible to ob-

tain satisfactory results on all the primates by using indirect methods of testing.

Distractibility and lack of motivation obviously hampered certain of the monkeys. The Cebus monkeys were extremely timid, although rather highly motivated, the guenons were easily distracted and one, "Thunder," was at times unmotivated. "Tommy," a baboon became irritable after the experiment had continued for about four months and was easily distracted thereafter. "Charlotte," the gibbon, seemed poorly motivated, and the macaque, "Sourface," was indifferent to the motive on certain days. The other animals were usually well motivated and not seriously distracted. Haggerty (14) had previously noted that primates become adapted to all ordinary disturbances of a Zoological park and can therefore be used as subjects for behavior tests. Haggerty (14), Boutan (7), and Watson (28) have reported the fact that a highly preferred food will serve as a strong incentive to the primate even when its diet is not curtailed.

Comparison of the genera

The results obtained on the comparison of the various genera agree very closely with the classification of the primates suggested by Tilney (26) on the basis of neurological evidence. Tilney suggests that on the basis of brain structure and relationships, the primates might be divided into four main groups, the lower primates, which would include the lemurs and the Platyrrhine monkeys; the intermediate primates, including all the Ceroopithecidae and the gibbon; the higher primates, the gorilla, the orang-outan, and the chimpanzee; and the human group or man.

There is little if anything to distinguish the results obtained by the lemur and the two Platyrrhine monkeys tested in this investigation and since they are definitely inferior to all others, they may be appropriately grouped as the lower primates on the basis of behavior as well as on the basis of brain structure.

On the basis of brain structure or brain indices Tilney finds no evidence for the superiority of the gibbon over the macaque nor of the inferiority of the baboon to either the macaque or the gibbon.

Our own results are confirmatory of these; indeed, in the animals tested the baboon appeared to be superior to both of the other genera.

Although our own results are too meager to make possible any effective comparison between the monkeys and the anthropoid apes, the work of other investigators makes it appear certain that the superiority of the brain structure of these primates as found by Tilney shows itself in increased capacity of response in the delayed reaction type of test.

Period of delay

On delays ranging from fifteen to one hundred and twenty seconds the best of our macaques did as well as or slightly better then the 2 macaques of Tinklepaugh, but we were entirely unable to obtain longer responses from our own animals, comparable to those obtained by Tinklepaugh. Although this may have resulted from inability to control factors of distraction and motivation as adequately as the above investigator, we believe that it is more probably due to the intrinsic nature of the problem itself, as Maslow obtained limits of delay of about the same magnitude even though the factors of distraction and motivation had been controlled. In Tinklepaugh's early experiments the containers were placed only 5 and 10 inches apart and the per cent of errors is high; later the cups were placed about 6 feet apart and the containers placed 6 feet from the animal. Under these conditions the per cent of errors decreased even when the period of delay was greatly extended. In our own experiments the inner sides of the cups were at all times 12 inches apart. The farther apart the cups and the greater distance they are away from the animal, the greater chance for association of the correct container with some other part of the situation. This factor may facilitate retention of the representative factor and thus explain the obtained differences. Further work on this problem is now being carried on by the writer. Such an explanation might also interpret the delays of from three to seventeen hours obtained under carefully controlled experimental situations by Adams (1) on cats, an animal definitely inferior to the primate in the phyletic series.

It is to be noted that our own results compare very favorably in the respect of maximum period of delay with those obtained by Allen (2) on one-year-old children, who was entirely unable to obtain responses when the period of delay exceeded sixty seconds.

VIII. SUMMARY AND CONCLUSIONS

1. Tests of delayed reaction on 24 primates indicate an increased capacity of response progressing from the lemur and South American monkey, to the old world monkeys, including the gibbon, and from these monkeys to the anthropoid apes. Among the old world monkeys the baboon appeared to be at least the equal of the macaque and the gibbon. These differences are indicated both by the maximum delays obtained and by the speed of the original learning. This capacity for adaptive behavior closely parallels the suggested classification of Tilney made on the basis of neurological researches.

2. The ability to solve tests of delayed reaction by means of representative factors appears as a common capacity of all the primates and varies only in degree.

3. Marked individual differences exist as well as differences between the genera. These may result from uncontrolled factors of age, motivation, emotional characteristics or innate differences in capacity.

4. There is evidence to indicate that, if the difficulty of the problem is such that the animal is unable to solve it immediately (insight), the primate characteristically attempts the solution in some simple and inadequate manner, gradually eliminates these methods, and finally solves the problem. If the difficulty of the problem is then increased, the animal tends to revert to the same inadequate types of response.

5. The direct method of delayed response is not solved by the primate on the basis of extraneous cues from the situation or from the experimenter. The reliability of the method is reasonably high, and once learned the method is retained with little loss for at least several weeks.

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